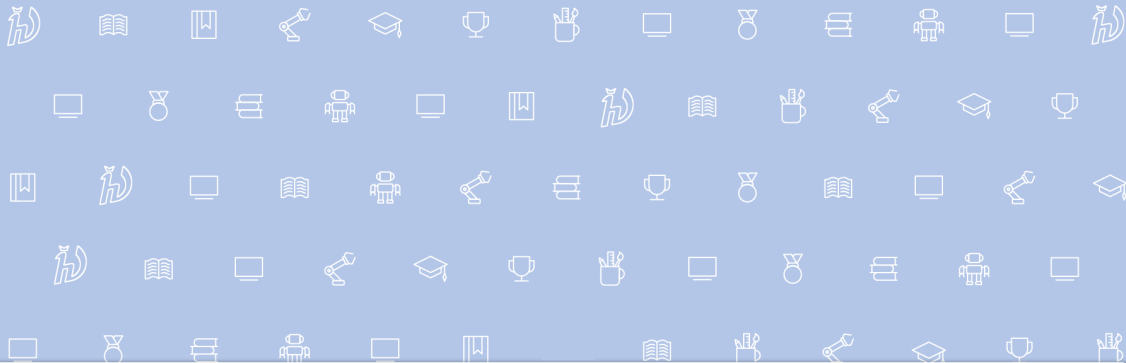




Artificial Intelligence Software, HW 7

Model Deployment with Triton



Center for
Cognitive
Modeling



Task 1 --- Model Preparation & Triton Deployment

Short description: Deploy a YOLO instance segmentation model to production using the full NVIDIA stack, following the tutorial below.

Artifacts: Repository with code and PDF report with screenshots for each step.

Requirements Tutorial & repo: https://youtu.be/ljqyuDxd_H0

https://github.com/Koldim2001/Triton_example

Use yolov8n-seg (instance segmentation) instead of the ResNet-152 shown in the tutorial.

1. **Export to ONNX:** export the model from PyTorch with a dummy input and dynamic axes so it accepts variable batch sizes.
2. **Triton file structure:** organize models/<name>/<version>/model.onnx and write config.pbtxt (platform, input/output tensor names, types, shapes, max_batch_size).
3. **Launch Triton in Docker** with GPU passthrough. Expose ports 8000 (HTTP), 8001 (gRPC), 8002 (metrics).
4. **FastAPI async service:** implement preprocessing identical to training pipeline, call Triton via tritonclient over gRPC asynchronously, return segmentation results (masks + class labels).



Task 1 --- TensorRT, Monitoring & Advanced Triton

5. **TensorRT optimization:** convert the ONNX model to .plan using trtexec inside the NVIDIA container. Enable FP16 quantization for maximum throughput on the target GPU.
6. **Monitoring:** configure Prometheus to scrape metrics from port 8002; import a Grafana dashboard showing RPS, latency breakdown (preprocessing / inference / postprocessing), GPU utilization, and average batch size.
7. **Advanced Triton features:**
 - Enable **dynamic batching** to merge concurrent single-image requests into one GPU batch.
 - Configure **instance group** to run multiple model copies in parallel.
 - Run a load test (e.g. Locust) and show Grafana metrics before and after enabling these features.

Deserve "A" grade!

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